Sleep, Physical Activity, and Acid-Related Upper-GI Disease

GERD and Oesophagitis in the *All of Us* Research Program (v9)

Analysis Specification, Software Architecture, and Verification

Functional GI Disorders & Physiological Measures Project

July 28, 2026

Abstract

This report specifies a retrospective cross-sectional analysis relating Fitbit-recorded *sleep* and *physical-activity* metrics to two nested electronic-health-record (EHR) outcomes — broad gastroesophageal reflux disease (`has_gerd`) and its erosive *oesophagitis* subset (`has_esophagitis`) — in the *All of Us* Controlled Tier CDR v9. The design is a 2×3 matrix: two outcomes crossed with three exposure sets (sleep, activity, and a combined set), giving six analysis configurations executed by one shared statistical engine. Seven R scripts implement the work without modifying any existing notebook. The analysis code has been written against the *verified* R9 schemas and validated end-to-end on schema-exact data; this document records the phenotype definitions, cohort logic, covariate strategy, statistical models, software architecture, and the verification results, including five defects found and fixed during schema alignment. Outputs are emitted in the exact table and figure format required by the target manuscript.

Code. `GERD Outcome Data Upload R9.R`, `GERD Data Prep R9.R`, `GERD Analysis Helpers R9.R`, `Sleep and GERD Analysis File R9.R`, `Activity and GERD Analysis File R9.R`, `Activity Sleep and GERD Analysis File R9.R`, `GERD Pharmacologics Data Upload.R`.

Contents

1 Overview and Research Question

Gastroesophageal reflux disease (GERD) sits at the intersection of two disease constructs: an acid-related structural disorder producing objective mucosal injury, and a symptom-defined presentation sharing mechanisms with the disorders of gut–brain interaction (visceral hypersensitivity, altered central pain processing, behavioural modulation). This dual character motivates analysing *two nested phenotypes* in parallel: broad, largely symptom-coded GERD, and the erosive oesophagitis subset with documented mucosal damage.

The question is directional: *sleep and physical activity* (exposures) → *GERD and oesophagitis* (outcomes). The prior hypothesis, consistent with the project's IBS, functional dyspepsia and constipation studies, is that greater sleep quantity/quality and higher activity are associated with *lower* adjusted odds of both outcomes, attenuating but persisting after adjustment.

1.1 Candidate mechanisms

These motivate both the covariate set and the directional predictions:

- **Acid clearance and supine time.** Swallowing and salivary bicarbonate are suppressed during sleep; fragmented or short sleep extends low-clearance recumbency and prolongs mucosal acid contact — a pathway that should matter *more* for the erosive phenotype.
- **Intra-abdominal pressure.** Higher BMI raises the gastro-oesophageal pressure gradient and promotes hiatal herniation, while also shortening sleep and reducing activity. This is the central confounder.
- **Lower-oesophageal-sphincter tone.** Nicotine, alcohol and sympathetic arousal increase transient sphincter relaxations.
- **Autonomic regulation.** Habitual activity raises vagal tone, which governs gastric motility and sphincter function; the corrected maximum-heart-rate exposure proxies this axis.
- **Visceral hypersensitivity.** Sleep deprivation lowers visceral pain thresholds, so identical acid exposure yields more symptoms — acting preferentially on the *symptom-coded* outcome. Contrasting the two outcomes therefore separates perceptual from structural pathways.
- **Reverse causation.** Nocturnal reflux itself fragments sleep. Adjustment cannot fix this; only design can, which is why a temporal outcome definition is the primary phenotype.

1.2 Evidence gap

Prior work measures sleep and activity largely by self-report and ascertains reflux by symptom questionnaire in modest samples. Few datasets link high-resolution wearable streams to longitudinal EHR at national scale. Notably, Hu et al. (2024) found subjective sleep quality differed sharply across the GERD spectrum while objective measures did not — a result that maps directly onto this study's two-outcome design. This analysis contributes objectively measured exposures, an EHR-defined outcome pair separating symptom-coded from erosive disease, and an unusually complete covariate set.

2 Study Design

3 Data Sources and Verified Schemas

Every column name and type below was verified by loading the dataset schema templates directly, not inferred. This matters: several assumptions that looked reasonable turned out to be wrong (Section 9).

3.1 Raw tables

3.2 Pre-built covariate files

The project already contains outcome-agnostic covariate objects produced by the published IBS and constipation pipelines. These are reused rather than recomputed (Section 5.4).

Element	Specification
Data source	*All of Us* Controlled Tier CDR v9 (`fc-aou-cdr-prod-ct.C2024Q3R9`)
Design	Retrospective cross-sectional; wearable exposures, EHR outcomes
Population	Fitbit-sharers, sharing EHR, age ≥ 18 at first Fitbit record
Exposures	Cohort quartiles (Q1 reference) of 8 sleep and 7 activity metrics
Outcomes	`has_gerd` and `has_esophagitis` (nested), ≥ 2 records each
Primary rule	≥ 2 records with the *first* ≥ 180 days after the first Fitbit record
Estimator	Logistic regression; e^{β_j} = adjusted OR for quartile j vs Q1
Multiplicity	Bonferroni (matching the antecedent IBS paper)
Configurations	Six: 2 outcomes $\times$ 3 exposure sets, one shared engine

Table 1: Design at a glance.

ID	Outcome	Exposure set	Cohort	Analysis file
C1	Broad GERD	Sleep (8)	≥ 180 nights	Sleep and GERD
C2	Oesophagitis	Sleep (8)	≥ 180 nights	Sleep and GERD
C3	Broad GERD	Activity (7)	≥ 30 days	Activity and GERD
C4	Oesophagitis	Activity (7)	≥ 30 days	Activity and GERD
C5	Broad GERD	Combined (15 + pairs)	Intersection	Activity Sleep and GERD
C6	Oesophagitis	Combined (15 + pairs)	Intersection	Activity Sleep and GERD

Table 2: The six configurations. Configurations sharing a row-pair are produced by the same file, because the driver loops both outcomes over one modelling frame.

3.3 Join hazards

Four schema features silently corrupt naive joins and are handled explicitly in the code:

1. `R9_avg_wear_hours_df` *and* `R9_steps_summary_filtered` both carry `n_valid_days`; joining them yields `n_valid_days.x/.y` and breaks every later reference. Only `person_id, avg_daily_wear_hours` is taken.
2. `alcohol_summary_df` carries `answer_concept_id.x` and `.y` from upstream joins; only `alcohol_likert_final` is taken.
3. `education_df`/`income_df` carry `answer_concept_id`. Categories are mapped from *concept IDs*, not the free-text `*_response`, whose values did not match the IBS script's string matching and silently collapsed everyone to "Missing".
4. Factor levels use **en-dashes** (`age_cat` "30–44"; `cci_cat` "1–2"). Dash style is normalised before use.

GERD is *not* present in `R9_condition_df`, so a dedicated outcome pull is required.

4 Outcome Ascertainment

Both outcomes use the project's $\geq$ 2-record specificity rule:

$$\texttt{has\_gerd} = \mathbb{1}\{\geq 2 \text{ GERD condition records}\},$$
$$\texttt{has\_esophagitis} = \mathbb{1}\{\geq 2 \text{ GERD records named ``esophagitis''}\}.$$

Because every oesophagitis record is also a GERD record, `has_esophagitis` is a strict subset of `has_gerd`; both are analysed as separate binary outcomes on the full cohort.

4.1 Concept expansion

A single BigQuery pull seeds two standard SNOMED concepts — **318800** (GERD) and **4223293** (GERD with esophagitis) — and expands to all standard, selectable descendants via the `cb_-`

Table	Rows	RAM	Columns used
`R9_fitbit_sleep_daily_summary_df`	31.8M	2.4 GB	`sleep_date` (Date), `is_main_sleep`, the 7 `minute_*`
`R9_fitbit_activity_df`	44.2M	2.4 GB	`date` (Date), `*_active_minutes`, `sedentary_minutes`
`R9_fitbit_intraday_steps_df`	34.2M	1.0 GB	`date`, `wear_hours`, `total_steps`
`R9_drug_df`	14.5M	3.1 GB	`drug_class_concept_id`, `drug_exposure_start_datetime`
`R9_survey_df`	9.7M	0.7 GB	`question_concept_id`, `answer_concept_id`
`R9_condition_df`	1.6M	274 MB	`condition_concept_id`, `condition_start_datetime`
`R9_measurement_df`	50,942	14 MB	`measurement_concept_id`, `value_as_number`
`R9_person_df`	59,018	5 MB	`date_of_birth` (character), race, ethnicity, sex

Table 3: Raw inputs. Date-like EHR fields are stored as *character* and require `as.Date()`; the Fitbit date fields are already `Date`.

File	Rows	Contents
`valid_population_sleep`	19,995	**the published cohort**: eligibility + age
`sleep_summary_filtered`	23,812	`n_valid_nights` + the 8 `avg_*` sleep metrics
`bmi_covariates_sleep_df`	46,753	`median_bmi`, `first_fitbit_sleep_date`
`cci_covariates_sleep_df`	11,652	`cci_score`, `cci_cat`
`medication_flags_sleep`	11,105	the 7 `on_*` flags
`comorbidity_status_sleep_ibs_df`	8,079	9 `has_*` (no `has_pud`)
`R9_steps_summary_filtered`	50,026	`avg_daily_steps`, `n_valid_days`
`R9_activity_zone_summary`	55,552	zone minutes incl. `avg_total_active_min`
`R9_comorbidity_status_df`	8,626	9 `has_*` *plus* `has_pud`
`R9_pud_status`	**137**	`has_pud` — very rare; see the sparsity guard

Table 4: Pre-built covariate inputs (sleep-anchored above, activity-anchored below the midline).

`criteria` hierarchy path match, restricted to Fitbit-sharers. Because the pull is already GERD-only, `has_gerd` is computed over *all* returned rows: re-filtering on the two seed IDs would discard every descendant-coded case and undercount GERD. The oesophagitis subset is identified within the same pull by a concept-name match, which is robust to release-to-release changes in descendant IDs.

4.2 Case definitions

The temporal rule is the primary phenotype because it is what the published IBS paper used, and because it is the only available guard against the reverse-causal edge (reflux → disturbed sleep) that covariate adjustment cannot address.

5 Cohort, Exposures, and Covariates

5.1 Eligibility thresholds

5.2 Exposures

Sleep metrics (minutes asleep, in bed, awake, restless, deep, light, REM, plus efficiency = asleep/in-bed) and activity metrics (daily steps, lightly/fairly/very/total active minutes, sedentary minutes, corrected maximum daily heart rate) are averaged within participant and then discretised into cohort quartiles with **Q1 as reference**. Quartiles avoid a linearity assumption,

Variable	Rule	Role
`has_gerd` / `has_esophagitis`	$\geq$ 2 records, first $\geq$ 180 days after the first Fitbit record	**Primary** — matches the published IBS paper's phenotype
`has_gerd_sens`	$\geq$ 2 records at any time	Sensitivity (always retained)
`has_gerd_dates`	$\geq$ 2 *distinct* dates	Specificity check
`severe_gerd_binary`	GERD *and* $\geq$ 2 post-diagnosis PPI/H_2RA dates	Optional treated-disease sensitivity

Table 5: Outcome variants. `GERD_PRIMARY_DEF` switches which rule is primary; the alternative is always kept as `*_sens`.

Rule	Threshold	Rationale
Main sleep only	`is_main_sleep`	Naps have different architecture
Valid sleep minutes	$(0, 1440]$	Excludes device/sync errors
Short-sleep exclusion	$<$ 30% nights <4h	Removes wear artifacts without discarding occasional short nights
Sleep coverage	$\geq$ 180 nights	∼6 months stabilises nightly means
Valid activity day	$\geq$ 10 wear h; 100–45,000 steps	Excludes non-wear and sensor spikes
Activity coverage	$\geq$ 30 valid days	Balances stability against sample size
Comorbidity rule	$\geq$ 2 distinct pre-Fitbit dates	Enforces temporality and specificity
Medication rule	$\geq$ 2 dates in prior 12 months	Sustained rather than incidental use

Table 6: Thresholds, inherited from the antecedent studies for comparability.

resist outliers, and reveal dose–response shape. Each exposure is modelled *separately* because exposures within a domain are collinear.

5.3 Combined design

The combined analysis uses the *intersection* cohort ($\geq$ 180 sleep nights *and* $\geq$ 30 activity days). It fits (A) each of the 15 exposures separately, and (B) mutually-adjusted models pairing one sleep with one activity exposure, so each domain is adjusted for the other:

$$\operatorname{logit}(p) = \beta_0 + \sum_{j=2}^{4} \beta_j^S \mathbb{1}[S{=}Q_j] + \sum_{j=2}^{4} \beta_j^A \mathbb{1}[A{=}Q_j] + \sum_k \gamma_k Z_k.$$

Four pairs are examined: minutes-asleep×steps, efficiency×very-active, deep-sleep×total-active, REM×sedentary. Covariate timing for this cohort is anchored to the sleep stream (always defined for its members).

5.4 Covariate strategy

The adjustment set matches the published IBS paper — age category, sex at birth, race, ethnicity, education, income, alcohol, smoking, median BMI, depression, anxiety, Charlson index — *plus* two GERD-specific gastrointestinal covariates, peptic ulcer disease (`has_pud`) and irritable bowel syndrome (`has_ibs`), both clinically entangled with reflux. Acid-suppression therapy is deliberately excluded from adjustment as a probable *mediator*; it is used only to define the treated-GERD sensitivity outcome.

Pre-built covariates are reused wherever they exist. This is a deliberate methodological choice, not merely an optimisation. The covariate derivations are outcome-agnostic and already underpin the published IBS paper; reusing them guarantees the GERD analysis is directly comparable to it, rather than differing through incidental re-derivation. It also avoids re-reading multi-gigabyte tables — `R9_drug_df` alone is ≈3.1 GB in memory, and loading the sleep, activity, steps and drug tables together approaches 9 GB. When a pre-built file is absent the covariate is derived from the raw tables instead, and the code reports which path it took.

6 Software Architecture

The implementation is layered so that all six configurations share one statistical engine and one data-loading path; only the cohort, exposure list and outcome column differ between them.

File	Role
`GERD Outcome Data Upload R9.R`	BigQuery pull of GERD records (both phenotypes) → `R9_gerd_outcome.rds`. Run once.
`GERD Data Prep R9.R`	**Data layer.** Schema-aware loading of cohort, exposures and the one-row-per-participant covariate bundle; prefer-pre-built-else-derive; NULL-safe joins; memory release after large tables are reduced.
`GERD Analysis Helpers R9.R`	**Engine.** Outcome definitions, exposure lists, adjustment set, labels; descriptive/univariable/multivariable routines; sparsity guard; diagnostics; the manuscript tables and figures.
`Sleep and GERD Analysis File R9.R`	Driver: C1 + C2
`Activity and GERD Analysis File R9.R`	Driver: C3 + C4
`Activity Sleep and GERD Analysis File R9.R`	Driver: C5 + C6, plus mutually-adjusted pairs
`GERD Pharmacologics Data Upload.R`	Optional PPI (21600095) / H_2RA (21600096) pull

Table 7: Code layers. Each driver sources the data layer and the engine, then calls one function.

```r
source("GERD Analysis Helpers R9.R")      # engine
source("GERD Data Prep R9.R")             # data layer

R9_gerd_outcome        <- read_first_existing("R9_gerd_outcome.rds", required = TRUE)
R9_gerd_outcome_status <- build_gerd_outcomes(R9_gerd_outcome,
                            first_fitbit_sleep_date_df, "first_fitbit_sleep_date")
covars_df              <- gerd_load_covariates(anchor = "sleep", ...)
# ... assemble one row per participant, create integer quartiles ...
modeling_df   <- prep_modeling_df(final_analysis_sleep_gerd_df, SLEEP_EXPOSURES)
sleep_results <- analyze_all_outcomes(modeling_df, exposures = SLEEP_EXPOSURES,
                                      coverage_var = "n_valid_nights", stub = "sleep")
```

Listing 1: A driver reduces to this.

Configuration lives in one place, at the top of the engine: `GERD_PRIMARY_DEF` (default `"post_fitbit"`), `GERD_POST_FITBIT_LAG_DAYS` (180), `GERD_P_ADJUST` (`"bonferroni"`), `GERD_MIN_CELL` (1), `GERD_OUTPUT_DIR`.

7 Statistical Analysis

7.1 Models

Descriptive tables are stratified by each outcome, reporting median [IQR] for continuous variables and n (%) for categorical, with t-tests and chi-square tests as in the antecedent paper. Univariable logistic regressions give unadjusted odds ratios for every exposure and covariate. For each exposure a separate adjusted model is fit,

$$\operatorname{logit}\big(\Pr[Y=1]\big) = \beta_0 + \sum_{j=2}^{4} \beta_j \mathbb{1}[X{=}Q_j] + \sum_k \gamma_k Z_k,$$

so e^{β_j} is the adjusted odds ratio for quartile j versus Q1. Reporting all three contrasts preserves the *shape* of the relationship rather than compressing it to a slope. Alongside each full model a backward-AIC model is fit with the exposure forced to remain (`MASS::stepAIC`), as a parsimony check: a stable estimate across both models indicates the association is not an artifact of one covariate set. Each model reports complete-case N, ROC AUC (`pROC`) and generalised VIF (`car`). Multiplicity is controlled by Bonferroni.

7.2 Separation and sparsity guard

Two features of this cohort make naive fitting dangerous: `has_pud` is recorded for only **137 participants** cohort-wide, and the oesophagitis outcome is itself uncommon. A covariate level containing zero events yields an infinite, meaningless odds ratio and destabilises the whole model.

The engine therefore screens every model before fitting. A covariate is dropped *from that model only* if it is constant, single-level, or has any empty outcome×level cell; each drop is reported and recorded on the result object. This converts a silent failure into an explicit, auditable decision. In validation the guard fired exactly as intended on the rare-outcome models, dropping `has_pud` and several sparse demographic factors. **If it fires on the primary models with real data, it must be reported in the manuscript's methods.**

```
guard      <- drop_unstable_covariates(d, outcome_col, covars, context = ...)
covars_use <- guard$keep       # constant / single-level / empty-cell terms removed
m_full     <- glm(as.formula(paste0(outcome_col, " ~ ",
                    paste(c(exposure, covars_use), collapse = " + "))),
                  data = d, family = binomial())
```

Listing 2: The guard, applied inside the per-exposure fitting function.

7.3 Diagnostics

Five assumption checks run for every configuration and are returned as numbers for the Results prose: independence (duplicate identifiers), multicollinearity (adjusted GVIF range), separation (large standard errors), influential points (maximum Cook's distance vs $4/N$), and linearity in the logit (Box–Tidwell for median BMI and continuous age).

7.4 Missing data and multiplicity

Binary EHR indicators are coalesced `NA`→`FALSE`: in EHR phenotyping the absence of a qualifying record means the participant does not meet the phenotype. Survey and measurement covariates are *not* imputed; an explicit "Unknown/Missing" level is carried through and reported, so differential missingness stays visible. Each adjusted model uses complete cases over its own variables, so the effective N is reported per model.

8 Manuscript Outputs

Running any driver writes, for *both* outcomes, into `./manuscript_output/`:

File	Manuscript element
`*_table1.csv`	**Table 1** — demographics by outcome group, median [IQR] / n (%)
`*_table2.csv`	**Table 2** — exposure averages interleaved with quartiles, cutoffs printed in the row labels
`*_supp_table1_univariate.csv`	**Supplement Table 1** — unadjusted ORs
`*_supp_table2_multivariable.csv`	**Supplement Table 2** — adjusted ORs, exposure rows, Q1 shown as "–"
`*_figure1_forest.png`	**Figure 1** — forest plots (Q1 reference)
`*_figure2_gvif.png`	**Figure 2** — GVIF multicollinearity chart
`*_diagnostics.csv`	GVIF range, max Cook's distance, Box–Tidwell p, AUC range
`*_quartile_cutoffs.csv`	Exact unrounded boundaries for the Table 2 footnote
`combined_*_mutually_adjusted_pairs.csv`	Sleep×activity pair estimates (C5/C6)

Table 8: Prefixes are `sleep_`, `activity_`, `combined_`, each followed by `gerd_` or `esophagitis_`.

Table 2 is emitted in the target format, with quartile boundaries computed from the data and written into the row labels:

```
Number of valid nights                    886 [532, 1,243]
Average sleeping minutes                    398 [369, 430]
Minutes asleep quartiles (minutes)
    Q1 (<369)                                   702 (25%)
    Q2 (369-398)                                705 (25%)
    Q3 (399-430)                                692 (25%)
    Q4 (>430)                                   693 (25%)
Daily steps quartiles (steps)
    Q1 (<5,785)   Q2 (5,785-7,489)   Q3 (7,490-9,108)   Q4 (>9,108)
```

Listing 3: Table 2 excerpt as produced by the code.

9 Verification

The analysis code was validated in three stages before release.

1. Schema validation. Every dataset the code touches was opened and its columns and types confirmed, rather than assumed — covering the raw Fitbit, EHR, survey and person tables and all pre-built covariate objects.

2. Schema-exact synthetic data. Synthetic datasets were constructed with column names and types checked field-by-field against the real schema templates, including the deliberate hazards: a duplicated `n_valid_days` column, the `answer_concept_id.x/.y` pair, en-dashed factor levels, and an ultra-rare `has_pud` (20 of 3,000, mirroring the real 137).

3. End-to-end execution. All three drivers were run to completion on that data. Final status: **exit code 0 for all three**, 50 output files, zero errors and zero warnings.

9.1 Defects found and fixed

Five defects surfaced during schema alignment. Each would have broken the run or silently corrupted results:

Defect	Consequence and fix
`library(MASS)` masked `dplyr::select()`	Every `select()` downstream failed immediately. MASS/car/pROC are now referenced with `::` and never attached; dplyr attaches last.
Duplicate `n_valid_days`	Joining the wear-hours and steps summaries produced `n_valid_days.x/.y`, breaking the coverage column. The wear file is now slimmed to two columns.
`answer_concept_id.x/.y`	Polluted the covariate bundle; only `alcohol_likert_final` is taken.
Education/income from free text	The `*_response` strings did not match the antecedent script's matching, silently collapsing every participant to "Missing". Categories are now mapped from *concept IDs*.
En-dashed factor levels	`cci_cat` levels failed to match and emitted 50+ warnings. Dash style is now normalised.

Table 9: Defects found by schema alignment and end-to-end execution.

9.2 Scope of the verification

This establishes that the code runs correctly against the real *schema* and emits the required format. It does not, and cannot, anticipate the missingness patterns and case counts of the real data. Two console outputs should be checked on the first production run: the eligible sleep cohort should print $N = 19{,}995$ (the published cohort; a different value means a different `valid_population_sleep` was picked up), and `build_gerd_outcomes()` prints case counts under both definitions — if the post-Fitbit count is very small, the "ever" rule may be more appropriate.

10 Sensitivity Analyses and Execution

1. **Alternative case definition** (`has_gerd_sens`): the "≥ 2 records ever" rule, always retained alongside the primary temporal rule.
2. **Distinct-date rule**: ≥ 2 records on distinct calendar dates, guarding against same-day code inflation.
3. **Sleep×activity overlap**: analyses restricted to concurrently well-measured days.
4. **Treated ("severe") GERD**: GERD *and* ≥ 2 PPI/H_2RA prescription dates after the first diagnosis — an EHR proxy for clinically managed disease, mirroring the IBS study's severe-IBS construction. Guarded by file existence, so the primary pipeline runs either way.

10.1 Execution order

1. Verify the GERD seed concepts in the Cohort Builder.
2. `source("GERD Outcome Data Upload R9.R")` $\rightarrow$ `R9_gerd_outcome.rds` (once).
3. Run any of the three drivers; each produces both outcomes.
4. Optionally run the pharmacologics pull, then re-run the sleep driver for the treated-GERD sensitivity.

All seven `.R` files and the `.rds` datasets must share one working directory. Detailed instructions, settings and troubleshooting are in `HOW_TO_RUN_GERD_ANALYSIS.md`.

11 Interpretation Framework

An adjusted OR of, say, 0.85 for Q4 versus Q1 of `min_asleep_quartile` means participants in the highest sleep-duration quartile have 15% lower adjusted odds of the outcome. A monotone decline across Q2→Q3→Q4 is a dose–response signature.

Exposure (high vs low)	Predicted aOR	Primary mechanism
More minutes asleep	< 1	Clearance, less arousal, lower sensitivity
Higher sleep efficiency	< 1	Consolidated sleep, less low-clearance recumbency
More deep / REM sleep	< 1	Restorative sleep, autonomic balance
More minutes awake / restless	> 1	Fragmentation, sympathetic arousal
More steps / active minutes	< 1	Weight control, vagal tone, motility
More sedentary minutes	> 1	Adiposity, postprandial recumbency
Higher corrected max HR	< 1	Cardiovascular fitness proxy

Table 10: Pre-specified directional predictions, applying to both outcomes.

Three consistency checks structure interpretation. *Across outcomes*: an association holding for both broad GERD and the erosive subset is less likely to be an artifact of symptom reporting; one confined to symptom-coded GERD points to a perceptual rather than structural pathway. *Across domains*: agreement between sleep-only, activity-only and mutually-adjusted estimates indicates neither domain is merely a proxy for the other. *Across case definitions*: agreement between the primary temporal rule and the alternatives indicates robustness to phenotype choice. Because the design is cross-sectional, estimates remain associations, not causal effects.

12 Limitations

- **Cross-sectional design.** The temporal phenotype mitigates but does not eliminate reverse causation; reflux itself disrupts sleep.
- **EHR phenotyping.** Diagnosis codes miss undiagnosed and over-the-counter–managed reflux and vary with coding practice; the ≥ 2-record rule trades sensitivity for specificity.

Participants with more health-care contact are both more likely to be coded and may differ in sleep and activity — partly blocked by adjusting for comorbidity and socio-economic status.

- **Name-based erosive definition.** Identifying oesophagitis by concept name may miss erosive concepts coded outside the GERD hierarchy unless their seeds are added.
- **Selection.** The cohort owns a Fitbit, shares it and the EHR, and wears the device enough to meet coverage thresholds — younger, more affluent and more health-engaged than the general population.
- **Residual confounding.** Diet, meal timing, caffeine, body position and hiatal hernia are unmeasured.
- **Precision.** Oesophagitis is rarer than broad GERD and the combined cohort is the smallest, so those configurations are the most precision-limited; the sparsity guard (Section 7.2) makes any resulting instability explicit rather than silent.
- **Covariate timing.** `has_pud` derives from a file computed on the activity-based anchor, a small documented inconsistency for the sleep analysis.

13 Conclusion

This specification defines a reproducible suite relating objectively measured sleep and physical activity to two nested EHR-defined outcomes in *All of Us* CDR v9. Six configurations are executed by a single engine over a single schema-verified data layer, so differences between them reflect biology rather than analytic drift. The code is written against verified schemas, reuses the covariate objects underpinning the published IBS paper for direct comparability, guards explicitly against separation in the rare-outcome models, and emits every table and figure in the target manuscript's format. Together with the temporal, distinct-date, overlap and treated-disease sensitivity analyses, the design supports a robust assessment of whether sleep and activity are associated with reflux disease and its erosive form.

A Appendix: Concept Reference

Purpose	Concept ID(s)	Note
GERD (outcome seeds)	318800; 4223293	Descendant-expanded
Oesophagitis subset	4223293 + name match	Erosive subset
BMI measurement	3038553	Body mass index
IBS (covariate)	75576; 4234788; 4261072; 4057826	General + subtypes
Peptic ulcer disease	4318534; 4134146; 4265600; 4027663	Via `R9_pud_status`
Depression / anxiety	440383 / 441542	
Diabetes / hypertension	201820 / 316866	
Heart failure / MI / stroke	316139 / 4329847 / 381316	
COPD / sleep apnea	255573 / 313459	
PPI / H_2RA	21600095 / 21600096	Drug-class ancestors
Beta-blockers / CCBs	21601664 / 21601765	
Stimulants / antidepressants	21604753 / 21604686	
Antipsychotics	21604490	
Anxiolytics	21604600; 21604565	
Hypnotics	21604635; 21604653; 21604685; 21604661	

B Appendix: Key Derived Variables

Variable	Meaning
`first_fitbit_sleep_date` / `first_fitbit_date`	Anchor for the temporal outcome rule and all pre-Fitbit covariate windows
`n_valid_nights` / `n_valid_days`	Coverage counts driving eligibility
`avg_min_*`, `avg_sleep_efficiency`	Participant-level sleep means
`avg_daily_steps`, `avg_*_active_min`, `avg_sedentary_min`	Participant-level activity means
`*_quartile`	Cohort quartile of the corresponding metric (Q1 reference) — the modelled exposures
`has_gerd` / `has_esophagitis`	Primary outcomes under `GERD_PRIMARY_DEF`
`has_gerd_sens`	The alternative case definition, always retained
`severe_gerd_binary`	GERD and ≥ 2 post-diagnosis PPI/H_2RA dates
`has_pud` / `has_ibs`	GERD-specific gastrointestinal covariates
`cci_score` / `cci_cat`	Charlson index, continuous and categorical
`median_bmi`	Within-person median BMI
`education_collapsed` / `income_collapsed`	Socio-economic factors mapped from concept IDs
`covars_dropped`	Covariates removed from a model by the sparsity guard
`R9_final_analysis_*_gerd_df`	The three one-row-per-participant modelling frames

End of report. This document describes code and methodology only; it contains no individual-level *All of Us* data.